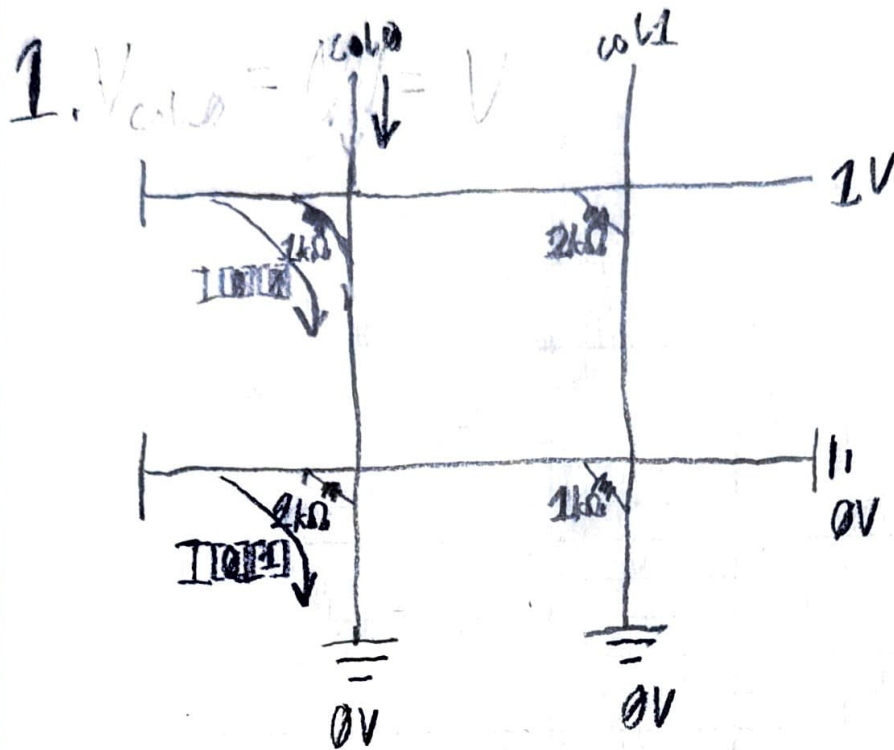
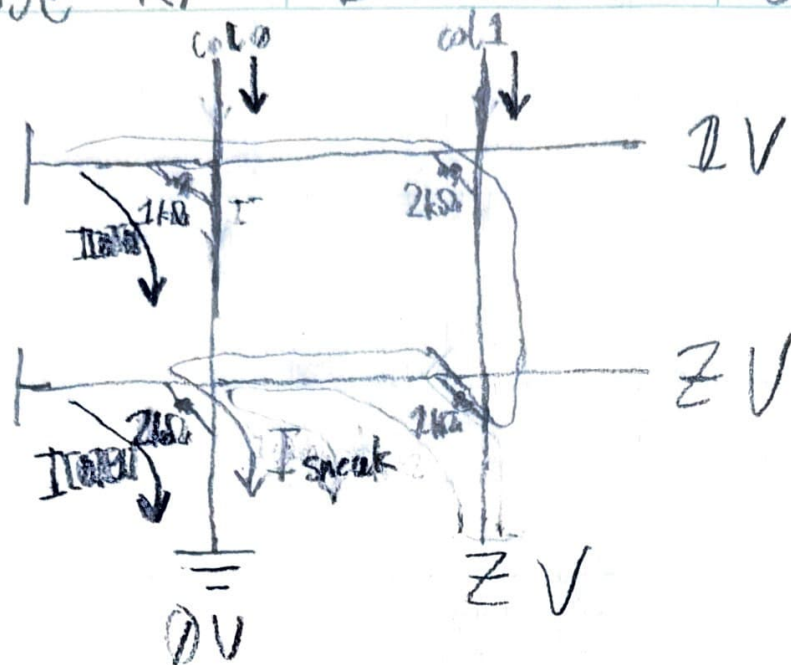




$$R = \frac{V}{I}$$

$$\begin{aligned}
 I_{col0} &= I[0][0] + I[0][1] \\
 &= \left(\frac{1V}{1k\Omega}\right) + \left(\frac{0V}{2k\Omega}\right) + (0V)(2k\Omega) \\
 &= \boxed{1mA}
 \end{aligned}$$

2.



$$I_{col0} = I[0][0] + I[0][1] + I_{sneak}$$

$$I_{sneak} = \frac{1V}{2k\Omega + 2k\Omega + 2k\Omega} = 0.2mA$$

$$I_{col0} = 1mA + 0A + 0.2mA$$

$$= 1.2mA$$

3. Current can flow bidirectionally across resistors allowing for different column currents to spill into one another. As you increase the number of the crossbar arrays, the number of current sneak paths increase and add to the ideal current output of any column.